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An IT Asset Governance Model Design Using COBIT 2019 And ITIL V4 Framework at BKU Itenas

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Abstract. Information technology is considered a critical asset in both institutions and large-scale companies. Assets within an institution/company support operational activities and business processes, making them closely related to IT governance, which directly involves IT services. Asset management in the General Finance Bureau (BKU) still faces challenges such as lack of human resources, undocumented software data, lack of audit implementation, and the absence of regular IT asset checks. COBIT 2019 with the BAI09 domain and ITIL V4 with ITAM practices are expected to work in synchronize to provide evaluation and recommendations regarding IT asset management at BKU Itenas. The research aims to measure the maturity level using COBIT 2019, conduct gap analysis, and design IT asset management governance using ITIL V4 in a sequential process starting from maturity analysis, gap analysis, design, and conclusions. The measurement of the maturity level obtained a value of 2.41 (managed), which is placed at maturity level 2. Meanwhile, target to be achieved at level 3 (defined). The GAP calculation between the current condition and the target to be achieved is 0.59. Furthermore, the recommendations include the creation of processes as guidelines for BKU Itenas in managing assets that support academic and administrative activities at Itenas.

1 Introduction

The application of information technology is an important element in the company because of the level of effectiveness and efficiency that accelerates every performance, so it affects the competition which is getting superior financially and network [1]. Information technology is used as an important asset in instances and large-scale companies. Assets in an institution/company support operational activities and business processes, this is aligned with IT governance which is directly related to information technology services. IT governance must be within the scope of the organization to align IT strategy and the strategy of the organization concerned by focusing on management activities and the use of information technology [2].

The Proper implementation of information technology requires an effective management of IT assets to ensure that business processes can run optimally. The lack of regular monitoring process and the lack of proper documentation of the monitoring results requires

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management to conduct meetings at least once a month to improve more effective communication between management and the IT asset management unit/section [3]. In addition, there are also problems regarding audits that have not been performed because if the audit is not performed immediately, the level of system maturity and gaps cannot be found [[1]4]. Problems regarding company assets occur because there is a lack of a good form of IT asset management in institutions/ companies. IT assets must be reliable to align IT Asset Management activities with business needs. IT Asset Management in an institution/company aims to plan, manage the asset life cycle, and manage possible risks that will occur in the future.

The proper implementation of information technology requires proper management of IT assets to ensure that business processes can run optimally. IT asset management in a system helps to optimize the IT asset control system to become more specific [5]. IT asset management cannot be separated from the work unit that is used as an important asset at ITENAS, which is BKU. BKU Itenas or referred to as the Itenas General Finance Bureau is an administrative unit responsible for managing financial and administrative aspects within Itenas. BKU has a focus in its implementation, which are financial management, budget planning, accounting, human resources, procurement, and other general services that are important in running operations efficiently. IT asset management must also consider the evaluation of IT service management to ensure that the work system in IT can be improved to achieve organizational goals [6]. Evaluation can be done using the ITIL (Information Technology Infrastructure Library) framework.

ITIL is a framework that provides IT service guidance in managing IT infrastructure, development, and operations. ITIL is often used as a reference in implementing ITSM / IT service management until it is properly integrated between IT and organizational strategy [7]. In this research, the latest version of ITIL, namely ITIL V4, is used. This research uses the latest version of ITIL, namely ITIL V4. The previous version of ITIL or ITIL V3 has a Service Asset and Configuration Management (SACM) best practice that focuses on service asset and configuration management. Service Asset and Configuration Management is the process of managing information about configuration items (CI) and the relationship between CI required for the delivery of an IT service. ITIL V4 provides the guidance needed by companies to solve new service management challenges and utilize IT potential in a flexible, well-integrated, and coordinated manner for service governance that supports information technology [8]. In ITIL V4 there are several domains, each of which has different processes, functions, roles, and performance. ITAM is a process of recording, documenting, and updating information about service assets related to IT services managed by service providers. When adjusted to the literature, ITIL V4 has a smaller and more specific scope so that it only focuses on IT Asset Management (ITAM). ITAM is a process of recording, documenting, and updating information about service assets related to IT services managed by service providers. [9]. ITAM has a role in maintaining IT assets in the form of computers and networks controlled/maintenance to control risks that may occur in the future. In supporting IT asset management, ITAM updates information containing asset specifications, the condition of each asset, mechanisms, relationships between assets, and proper utilization of assets.

In addition to ITIL V4 with IT Asset Management best practices, the COBIT 2019 framework with the domain used in this research is BAI (Build, Acquire, and Implement) with the BAI09 (Manage Assets) sub domain. The use of COBIT 2019 with the BAI09 domain is conducted as a guide in managing IT assets through their life cycle so that their use provides maximum value to the organization [10]. BAI09 is used in research to measure how well institutions/companies perform processes in managing IT assets. The effectiveness of IT asset management in institutions/companies is based on maturity level as a benchmark to determine performance at each level of the focus area. Measurement is needed to find out the activities performed by the company to produce more effective management and find out the various processes based on the ITIL guiding principles in the start where you are section. Guiding principles are universal principles that direct institutions/companies in all situations so that they are not fixed on their goals, strategies, type of company, and management structure. The start where you are principle considers everything that has been previously available in the form of services, processes, programs, projects, and workers to create better results. After that, the GAP analysis process can be used to compare the current situation with all the needs and goals that the institution/company wants to achieve through the gap/GAP in each management practice.

The purpose of this research is to measure the maturity of IT asset management governance using COBIT 2019 at BKU Itenas, measure the GAP analysis of the maturity level results obtained at BKU Itenas, and design IT asset management governance using ITIL version 4 at BKU Itenas. COBIT 2019 with the BAI09 domain and ITIL V4 with the ITAM practice are expected to work in harmony in providing evaluations and recommendations for the management of IT assets contained in UPT-TIK Itenas so that research is carried out related to IT Service Governance at BKU Itenas by the author using ITIL V4 and COBIT 2019.

2 Materials and Method

2.1 Comparison of Frameworks

According to Fig. 1, COBIT has a balanced approach between horizontal and vertical dimensions that compares favorably with other standards. These dimensions map IT processes into specific objectives. According to Kaban [11], COBIT has a wider and more detailed spectrum of IT processes. COBIT covers most of the IT processes that must be managed in an organization more completely and specifically. ITIL has detailed guidance on implementing and managing IT processes that address technical and operational aspects, while COSO has fewer details with a broad operational scope.

2.2 ITSM

IT Service Management (ITSM) is management that has the aim of providing services to support the achievement of the goals of an organization / business working together to ensure and monitor the quality of IT services that have been agreed upon by customers by focusing on the level of alignment of information technology services effectively and efficiently. [12].

2.3 COBIT

COBIT (Control Objectives for Information and Related Technology) is a framework that provides standards and guidelines for IT activities, with the aim of optimizing investments using information technology and ensuring effective service delivery by providing a clear matrix for managing errors [13]. In this study, an analysis was carried out using the Build, Acquire, and Implement (BAI) domain in the BAI09 process (Managing Assets). BAI domain measures the level of IT capability to prevent the occurrence of IT Productivity Paradox. IT Productivity Paradox is an IT adoption in an organization that increases productivity significantly and indirectly.

2.3.1 Process Capability Level

Capability levels that start from 0 to 5 are obtained from each governance and management objective that operates within a certain level, ranging from 0 to 5 as described in Table 1 [15]. The capability level is used as a benchmark for how well a process is implemented and performed [16].

Table 1. Process Capability Level

Level	Description
0	The process lacks basic capabilities and reflects an incomplete approach to addressing, governance and management objectives. It may or may not fulfill the intent of any process practice.
1	The process more or less achieves its objectives through the application of an incomplete set of activities that can be characterized as preliminary or intuitive and not very organized.
2	The process achieves its goal through the application of a basic, yet complete, set of activities that can be characterized as already done.
3	The process achieves its objectives in a much more structured way using the assets that the organization has. The process is usually well defined.
4	The process achieves its objectives, is well defined and its performance is measured quantitatively.
5	The Processes achieve their objectives, are well defined, their performance is measured to improve performance and continuous improvement is sought.

2.3.2 Activity Process Assessment

Capability levels can be achieved in varying degrees based on ratings [17]. The available ratings depend on the context in which the performance assessment is made [18].

Table 2. Activity Process Assessment

Rating Levels		
N	Not achieved	Capability level achieved less than15%
P	Partially achieved	Capability level achieved more than 15% to 50%
L	Largely achieved	Capability level achieved more than 50% to 85%
F	Fully achieved	Capability level achieved more than 85% to 100%

2.3.3 Focus Area Maturity Levels

A higher level is required in knowing the applicable performance for each process capability rating [16]. Maturity levels can be used for the purpose of knowing the applicable performance.

Table 3. Focus Are Maturity Levels

Level	Description
0 (Incomplete)	Work may or may not be completed to achieve the governance and management objectives of the focus area
1 (Initial)	Work is complete, but the full purpose and intent of the focus area has not been achieved.

2 (Managed)	Performance planning and measurement are in place, although not yet in a standardized way.
3 (Defined)	Company standards provide guidance across the company.
4 (Quantitive)	The company is assisted by data, with quantitative performance improvement.
5 (Optimizing)	The company is focused on continuous improvement.

2.3.4 Maturity Level Measurement

Maturity level measurement is carried out based on the results of the questionnaire. The results of the questionnaire will be recapitulated so that it can produce a comprehensive capabilty level of each BAI09 domain management practice activity, then a maturity level value is obtained from the results of the BAI09 domain capability level.

If the data from the results of filling out the questionnaire has been obtained, these results can be calculated to get the maturity level value with the following formula:

1. **Questionnaire Value Conversion.** The questionnaire value conversion scale applied in this study is shown Table 4.

Table 4. Score Index

Index	Score
N	0
P	0.33
L	0.66
F	1

2. **Average Conversion.** The converted scores can be calculated using the following formula:

$$coversion = \frac{conversion\ score}{\sum question} \tag{1}$$

3. **Normalization.** The conversion value that has been summed up from each level will be divided by the average conversion value of the entire level and multiplied by the level so that the normalization value is obtained. The calculation of the normalization value can be expressed by the following formula:

$$normalization = \frac{conversion}{\sum level} \times Capability\ level \tag{2}$$

4. **Maturity Calculation.** The calculation of the maturity value is obtained by adding up the normalized value of the entire level multiplied by 2, then the average maturity value can be done by adding up the total maturity value and then dividing by the number of respondents. The formula used can be expressed as follows:

$$Maturity\ Level = \sum Normalization \times 2 \tag{3}$$

$$maturity\ level = \frac{\sum Maturity\ level \times Capability\ level}{\sum Respondent} \tag{4}$$

The score calculation will produce a final value, then the maturity level will be known based on the range of values shown in Table 5.

Table 5. Maturity Score

Score Range	Maturity Score	Level Maturity
0 – 0.50	0	Incomplete
0.51 - 1.50	1	Initial
1.51 – 2.50	2	Managed
2.51 – 3.50	3	Defined
3.51 – 4.50	4	Quantitative
4.51 – 5.00	5	Optimizing

2.4 ITIL V4

ITIL V4 is the latest version of ITIL with updates to many ITSM methodologies in a broader context taking into consideration customer experience, value streams, and digital transformation, as well as adopting new ways of working such as Lean, Agile and DevOps. Although more specifically a service management guide, ITIL V4 is often used as an IT administration guide.

2.4.1 Four Dimensions of Serivice Management

ITIL 4 contains the provision of guidance that organizations need to handle service management with various challenges and potential utilization of modern technology called the 4 Dimensions of Service Management (Fig. 1).

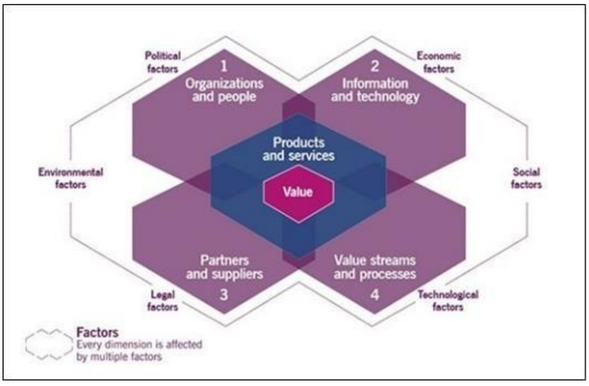

Fig. 1. 4 Dimensions of Service Management

The 4 Dimensions of Service Management aims to ensure a flexible, coordinated, and integrated system for IT service governance and management. The 4 Dimensions contained in Fig. 2. Consists of Organizations and People, Information and Technology, Partners and Suppliers, and Value Stream and Processes. In the application of governance with ITIL V4 carried out by the Service Value Chain as the implementation life cycle in the form of Service Value Stream [19].

2.4.2 IT Asset Management

IT Asset Management is implemented with the aim of planning and managing the lifecycle of all IT assets. When the company needs management, IT Asset Management will maximize value, control costs, manage risks, support decision making about the purchase, reuse, retirement, and disposal of assets that have met the terms and contracts that have been established [19].

The contribution of IT asset management to the service value chain, with practices applied mainly to design and transition, and obtain/build service value chain activities with the following details:

1. Plan. Some asset management policies are driven by governance, and some are driven by other practices, such as information security management. IT asset management is considered a strategic practice that benefits the organization for understanding, cost management, and value management.
2. Improve. The impact of chain activities must be considered on IT assets and some improvements directly involve IT asset management to help manage costs.
3. Engage. Requests for IT Asset Management are made by stakeholders, for example a user may report a lost/stolen cell phone, or a customer may require a report on the value of IT assets.
4. Design and Transition. This value chain activity is natural to change the status of IT assets to drive most IT asset management activities.
5. Obtain/Build. IT asset management supports asset procurement so that assets can be tracked from the beginning of their life cycle.
6. Deliver and Support. IT asset management helps locate IT assets, track their movement, and control their status within the organization.

3 Results

3.1 Questionnaire Analysis Results

The results of the questionnaire calculations that have been carried out are shown in Table 6.

Table 6. The Results of Maturity Level

Domain	Respondent	Level 1	Level 2	Level 3	Level 4	Level 5	Maturity Score
BAI09	R-01	0.53	0.93	0	0	0	2.82
	R-02	1	0	0	0	0	2
	Average						2.41

After calculating the results of the questionnaire, it is known that the BKU gets an average value of 2.41 which is included in maturity level 2, Managed in managing IT Assets. This means that BKU has designed and implemented IT asset management at Itenas even though it is not in accordance with the standards.

3.2 Determination of Maturity Target

Determination of maturity expectation conditions is carried out at this stage based on the BAI09 Managed Assets Domain using a questionnaire filled out by 1 respondent.

Table 7. Determination of Maturity Target

Focus Area and Current Maturity	Target Maturity Level	Situational Consequences of Maturity Level	Responds	Researcher's Rational Analysis

IT Asset Management (Level 2, Managed)	Level 3, Defined	At this level, BKU Itenas relies on standards as guidelines and guidance in managing IT assets. At Level 3 of the BAI09 process, the target is that all activities should achieve a grade of "L" or "F".	Yes / No	Currently, BKU Itenas is in a condition where it must be able to adjust and develop IT asset management within Itenas. The goal is that BKU Itenas is considered to have the ability to implement Maturity Level 3, namely "Defined".
	Level 4, Quantitative	At this level, BKU Itenas seeks to improve the performance of IT assetmanagement using data and other resources. At Level 4 of the BAI09 process, the target is that all activities should achieve a grade of "L" or "F".	Yes / No	The results of observations and interviews conducted by researchers show that currently BKU Itenas still faces data and resource limitations in managing IT assets. This is due to the absence of guidelines that have been reviewed to regulate the process.

3.3 Gap Analysis

After knowing the average maturity value at the previous stage, the results will be compared and then the gap value will be obtained as shown in Fig. 2. The target owned by BKU Itenas is the development of service quality and performance based on standards that serve as guidelines for managing IT assets. This indicates that the target to be achieved by BKU is at level 3 (Defined) or equivalent to maturity level 3.

Table 8. GAP Maturity Level

No	Domain Proses TI		Maturity Level		
			Current	**Expectation**	**GAP**
1	BAI09	Managed Assets	2.46	3.00	0.54
Total			2.46	3.00	0.54
Average			2.46	3.00	0.54

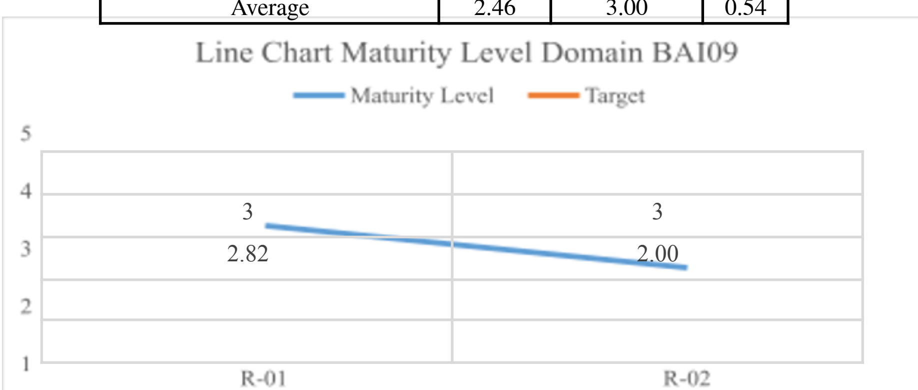

Fig. 2. Chart GAP Analysis

3.4 The 4 Dimension of Service Management

The 4 dimensions of service management by BKU are used to recognize and pay attention to important elements in IT service management. This identification helps in forming a comprehensive and overarching framework for managing IT services, with a focus on delivering value to customers and achieving business objectives as follows.

3.4.1 Organizations and People

The targets and roles that need to be achieved by each related party at Itenas in managing IT assets with the addition to consider including additional staff as described in Table 9.

Table 9. Targets and Roles that need to be achieved.

Role (ITIL V4)	Role (BKU)	Analysis	Status
IT Asset Manager	Head of Sub-Section Procurement and Inventory Responsible for overseeing the lifecycle of IT assets	Based on the researcher's analysis, the role of the Head of Sub-Section Procurement and Inventory is responsible for overseeing the life cycle of IT assets with organization-related data.	Available
IT Asset Custodian	Not available Responsible for IT asset lifecycle care or maintenance.	Based on the researcher's analysis, the BKU does not yet have a role that is responsible for maintenance or maintaining the asset's life cycle, so there are several activities that cannot be carried out.	Not available
License Manager	Not available Understanding the software and cloud license aspects is not available in BKU.	Based on the researcher's analysis, the BKU does not yet have a role that understands software and cloud licenses, so it needs to be added in managing IT assets.	Not available
Contract Manager	Head of General Section Responsible for asset contracts.	Based on researcher analysis, the role of the Head of the General Section is capable of handling contracts related to IT assets.	Available
IT Coordinator	Head of Sub- Section Procurement and Inventory Responsible for coordinating and operational management.	Based on researcher analysis, the role of the Head of the Procurement and Inventory Subdivision is capable of carrying out its duties in coordinating IT assets in the organization.	Available
IT Asset Owner	Dayang Sumbi Foundation Asset owner or full power of asset ownership.	Based on researcher analysis, IT Asset Owner parties are not added because ownership at Itenas is fixed.	Available
IT Asset Consumer	Study Programs and Work Units Parties that use assets at Itenas.	Based on researcher analysis, IT Asset Consumer parties are not added because consumers at Itenas are permanent.	Available

3.4.2 Information and Technology

The following are recommended tools to achieve maturity level 3, there are tools used to fulfil the activities in Table 10.

Table 10. Recommendation Tools

Tools	Activities	Recommendation
Service Now	The BKU has not assessed the maintenance associated with assets, including the cost of repairs and replacement parts.	ServiceNow provides various reporting and analysis tools to generate reports that help in making better decisions regarding IT assets.
	The BKU has not used statistical data related to asset capacity and usage to identify assets that are not optimally utilized or are redundant.	ServiceNow delivers analytics, reporting, and dashboard features that help track IT asset performance and provide insights into overall asset management.

3.4.3 Partner and Suppliers

The following are the steps recommended by ITIL V4 based on IT asset management practices:

1. Sign confidentiality agreements and control the exchange of sensitive IT asset data with suppliers and partners: Confidentiality agreements (NDAs) and strict controls must be implemented to protect the confidentiality of IT asset data in relationships with suppliers and partners.
2. Finding conflicts of interest: Organizations should be alert to conflicts of interest with suppliers or partners, such as fees based on the amount of assets provided or the existence of conflicting interests by suppliers who are also resellers or auditors.
3. Include provisions in third-party service contracts on handling IT assets with clear duties and responsibilities, such as receipt of ordered IT assets, secure storage, stock management, allocation/IMAC, detection and handling of special situations, inventory, and more.
4. Communication of achievements A mature ITAM improves relationships with vendors and suppliers. For example, understanding software licenses and compliance avoids time-consuming audits. IT expenditure control focuses on value-added cooperation with suppliers. By running efficient operations, organizations can attract talented workforce interested in working with BKU and reduce service provider costs.

3.4.4 Value Stream and Processes

Value streams are a series of interconnected and coordinated activities to create, deliver, and deliver value to end customers. Meanwhile, processes are a series of steps that have been structured and well organized to achieve certain goals or produce desired outputs. The following is a mapping of Value Stream and Process.

1. Software License Documentation Value Stream
   The software and license documentation process on the Itenas asset website is an important process in ensuring compliance with the law, providing transparency about the use of software, and copyright related to the Itenas BKU, by selecting the practices needed to create an effective software and license documentation process.
   The value stream was obtained from the request of the asset manager who wanted to document the IT software license in the future, then the researcher further identified the opportunity. Fig. 3 shows the specific value chain flow in adding software and license documentation features. The value chain flow is obtained from the value stream and process because it takes a collection of activities to convert inputs into outputs. Value stream activities of software documentation and license proposed in this study are triggered by demand or requests to create new features from managers.

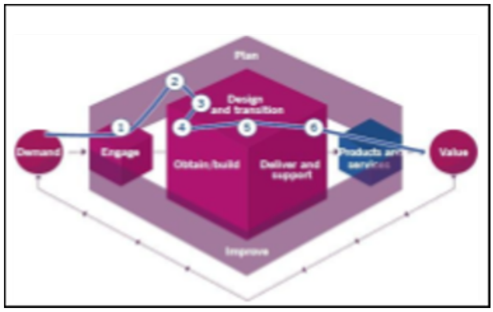

Fig. 3. Service Value Stream of Software License Documentation

There are service value chain activities according to ITIL V4 which includes: (1) engage to receive requests from managers in the documentation process; (2) plan to decide on adding software documentation and license features; (3) design and transition in order to design the software documentation and licenses features; (4) Obtain/build: considering that this service has never existed before, so that the activity related in this stage is to obtain and build the new feature in existed system. After the development and testing stages, the service may need improvements so that activities return to; (6) Design and Transition activities; (6) Deliver and Support: and the last activity is delivering the new feature and support the user while using the new feature.

2. **Regular Maintenance of IT Assets Value Stream**

According to gap analysis, BKU does not yet have a role that is responsible for maintenance or maintaining the asset's life cycle, so there are several activities that cannot be carried out. Therefore, we proposed regular maintenance of IT Assets Value Stream in this study which is triggered by demand or requests to create new services.

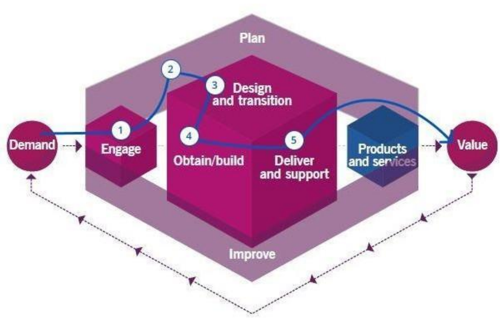

Fig. 4. Service Value Stream of Regular Maintenance of IT Assets

There are 4 activities according to ITIL V4. These activities are consists of (1) Engage: to receive requests from users in the IT asset maintenance process; (2) Plan: to determine the scheduled maintenance policy for IT assets; (3) Obtain/build: Activity related to this stage is obtain or acquire tools to help maintain IT assets in organization; (4) Design and transition: to implement the policies that have been made, and (5) finally Deliver and Support: for documentation of the results of IT asset maintenance.

3. **Audit verification process Value Stream**
   The value stream was triggered by a request from the asset manager who expressed a desire in the future to verify and audit IT assets.

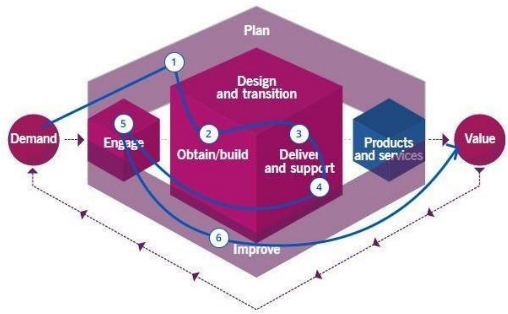

Fig. 5. Audit verification process

Researchers then further identified the opportunity. Fig. 5 shows the service value chain flow in the verification and audit of IT assets, the value stream starts from the manager's request who wants to audit the assets owned, then a service value chain flow is made according to ITIL V4 in which there are engage, plan, obtain/build, improve, deliver and support carried out twice that the audit has produced the value expected by the manager.

4 Conclusion

In this research used COBIT 2019 and ITIL V4 frameworks, where COBIT 2019 has guidelines on performance measurement and ITIL V4 provides recommendations on aspects that need to be improved to provide value in managing IT assets. Based on the results of designing an IT asset governance model using the COBIT 2019 and ITIL V4 frameworks, the following conclusions are obtained measurement of maturity level using COBIT 2019 in the BAI09 domain related to IT Asset Management at BKU Itenas, obtained a maturity value of 2.41 in Table 7, which places it at maturity level 2, which is managed. BKU determines the target to be achieved, namely, maturity level 3, defined based on consideration of the capabilities possessed by BKU. The determination of the maturity level target is used as a comparison with the maturity level obtained, where the results of the calculation produce a GAP between the current condition and the target that needs to be achieved of 0.59 in Table 8. The results of the GAP analysis can be developed into recommendations on governance in IT asset maintenance, IT asset documentation, and IT asset verification and audit at BKU Itenas. Therefore, in developing IT asset management governance at BKU Itenas, the ITIL V4 framework with a value stream approach is used. This approach contains a series of service value chains that help identify steps, processes, policies, and standards for IT asset maintenance, IT asset documentation, and IT asset audits at BKU Itenas. In addition, the design recommendations also include creating a process as a guide and guideline for BKU Itenas in managing related assets that support the implementation of education at Itenas.

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